

1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

- After starting a capture of Instagram.com website on Wireshark. Here, these protocols have been used in the packet list.

Protocol	Purpose
DNS	Resolves the Instagram.com into an IP address.
ARP	Finds the MAC address of devices on the local network.
TCP	Establishes reliable network connections
TLSv1.2/TLSv1.3	Encrypts the communication between browser and the website (HTTPS).
HTTP/2 or HTTP	Web traffic
QUIC	Used by many modern browsers for faster HTTPS over UDP.
UDP	Transport protocols used by DNS and QUIC.
ICMP	Network diagnostic messages.

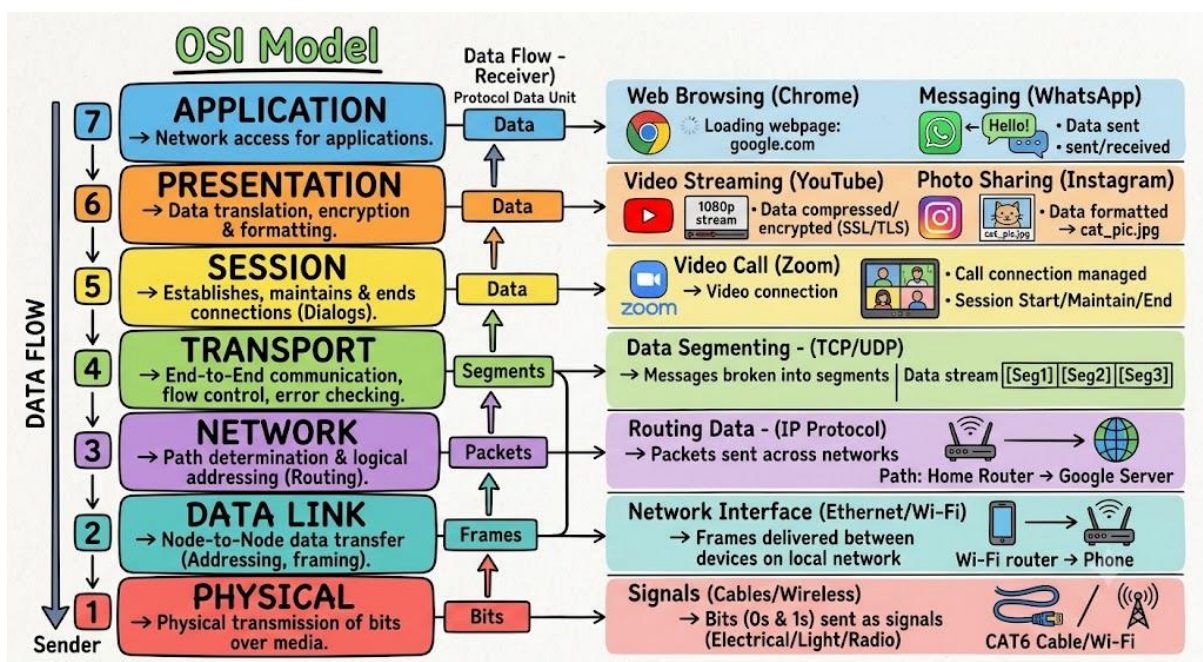
2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

- After capturing packets when I visited to Instagram.com. Here are some packets which mainly represents OSI layer.

Packet No.	Protocol	Main OSI layer	Reason
1	DNS	Application layer (Layer-1)	DNS translates the Instagram.com into the IP address.
2	TCP	Transport layer (Layer-4)	TCP establishes a reliable connection between computer

			and the web server using three-way handshake.
3	TLS or HTTPS	Application layer (Layer-1)	TLS encrypts the HTTP communication between browser and website keeping data secure.

3. Create a colourful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).



4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

- When you send a message on WhatsApp. It passes through many OSI model layers before reaching the other person. Here's a simple explanation using 4 OSI layers:

1. Application Layer (Layer-1):

- This is where we type a message in WhatsApp and press Send. WhatsApp prepares the message and starts the sending process.

2. Transport Layer (Layer-4):

- The message is broken into smaller pieces and it's called as a segments. This layer makes sure all parts of the message are delivered correctly and in the right order using TCP protocol.

3. Network Layer (Layer-5):

- This layer adds the IP address of the sender and receiver. It decides the best path for the message to travel across the internet from phone to the receiver's phone.

4. Data Link Layer (Layer-6):

- The message is placed into frames and sent over Wi-Fi or mobile data. This layer uses the device's network hardware to send the data to the router or mobile network.